19 For all values of x

$$f(x) = (x + 1)^2 \quad \text{and} \quad g(x) = 2(x - 1)$$

(a) Show that $gf(x) = 2x(x + 2)$

$gf(x) = g(f(x)) = 2((x + 1)^2 - 1)$

1 mark
Put f(x) into the input of g

	x	1
x	x^2	x
1	x	1

$= 2(x^2 + 2x + 1 - 1)$

$= 2x^2 + 4x = 2x(x + 2)$

1 mark
Expand, simplify and factorise

(2)

(b) Find $g^{-1}(7)$

To find $g^{-1}(7)$, solve $g(x) = 7$.

$$2(x - 1) = 7$$
$$\div 2 \qquad \div 2$$
$$x - 1 = 3.5$$
$$+1 \qquad +1$$
$$x = 4.5$$

1 mark
Reverse the operations of g

Check: $2(4.5 - 1) = 7$.

1 mark
The inverse input is 4.5

4.5

(2)

(Total for Question 19 is 4 marks)